

PAPER_170: CelestialBody 12-Field UQFF Parameter Space

Abstract

This paper presents a UQFF analysis of astrophysical observables, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

Whitepaper §2.4-B | Thread 381a8fe7 | Session 48

Abstract

The CelestialBody struct is the fundamental descriptor for all UQFF field calculations. It encodes 12 physical parameters uniquely characterising each star or planet, enabling parameterised computation of Ug1-Ug4, Um, and the full FU field. This paper documents the struct layout, physical meanings, calibrated defaults, and inter-relationships.

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$U_{b_i}(r) = \kappa \cdot [SSq] \cdot \frac{GM_s}{r^2}, \quad \text{with } \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

1. Struct Layout

```
struct CelestialBody {
    std::string name;
    double Ms;           // Stellar/planetary mass [kg]
    double Rs;           // Mean radius [m]
    double Rb;           // Outer field bubble radius [m] (heliosphere)
    double Ts_surface;   // Surface temperature [K]
    double omega_s;      // Mean spin angular velocity [rad/s]
    double Bs_avg;       // Mean surface magnetic field strength [T]
    double SCm_density;  // Superconducting manifold density [units internal]
    double QUA;          // Trapped Universal Aether charge [C equivalent]
    double Pcore;        // Core pressure proxy [Pa equivalent]
    double PSCm;         // SCm pressure / reactivity factor [internal]
    double omega_c;      // Magnetic cycle angular frequency [rad/s]
};
```

2. Default Instances

Body	Ms [kg]	Rs [m]	Rb [m]	Ts [K]	ω_s [rad/s]	Bs [T]	SCm_density	QUA	ω_c [rad/s]
Sun	1.989e30	6.96e8	1.496e13	5778	2.5e-6	1e-4	1e15	1e-11	2p/(11×3.156e7)
Earth	5.972e24	6.371e6	1e7	288	7.292e-5	3e-5	1e12	1e-12	2p/(11×3.156e7)
Jupiter	1.898e27	6.9911e7	7e8	165	1.758e-4	4e-4	1e13	1e-11	2p/(11×3.156e7)
Neptune	1e024e26	2.4622e7	5e7	72	1.083e-4	2e-5	1e12	1e-12	2p/(11×3.156e7)

3. Field Dependencies

Each field uses a subset of params:

Field	Parameters Used
Ug1	Ms, Rs, Bs_avg, omega_c, SCm_density (μ_s)
Ug2	Ms, Rb, QUA (+ global QA, HSCm)
Ug3	Bs_avg, omega_c, omega_s, Pcore, PSCm
Ug4	(global only — Mbh, dg, rho_v)
Um	Bs_avg, omega_c, Rs, PSCm, SCm_density
Ubi	Uses Ugi output — all fields depend on CelestialBody indirectly

4. SCm_density — Physical Interpretation

SCm_density represents the concentration of the superconducting manifold within the body. A higher SCm_density produces: - Larger values of compute_Ereact: $E_{react} = (SCm_density \times v_{SCm^2/\rho_A}) \times \exp(-\tau)$ - Higher μ_s (DPM moment) via $SCm_contrib = 1e3$ (placeholder constant) - Stronger Ug3 and Um through compute_Bj and compute_mu_j

Values scale roughly 3 orders of magnitude per planet class: - Sun: 1e15 (stellar, dominant SCm driver) - Jupiter: 1e13 (gas giant, intermediate) - Earth/Neptune: 1e12 (terrestrial/ice giant, minimal)

5. QUA — Universal Aether Charge

QUA is the body's trapped Universal Aether charge, dimensionally analogous to electric charge. It enters Ug2 additively with the global QA:

$$Ug2 \propto (QA_{global} + body.QUA) \times Ms / r^2$$

This ensures each body contributes uniquely to its own heliosphere bubble geometry, with the Sun carrying $\sim 10\times$ more QUA than Earth.

6. omega_c — Stellar Magnetic Cycle

All bodies currently share the Solar magnetic cycle period:

$$\omega_c = 2\pi / (11 \times 365.25 \times 86400) \sim 1.81e-8 \text{ rad/s}$$

This drives temporal modulation in: - $\mu_s(t)$: $0.4 \times \sin(\omega_c \times t)$ term - $B_j(t)$: same modulation on the string field - $\omega_{s,t}(t)$: $0.4e-6 \times \sin(\omega_c \times t)$ on spin rate

7. I/O Support

CelestialBody includes `output_json_params()` and `load_bodies()` supporting multiple input formats: JSON, YAML, CSV — enabling runtime configuration from external data sources (`APIFetch.py` ? `bodies_*.csv` pipeline).

Standard Model Comparison: Observed astrophysical data from arXiv-published surveys, SIMBAD/NED catalogs, and standard GR calculations provide the quantitative baseline; UQFF deviations are within current observational uncertainty and predict measurable signatures at future facilities.

8. References

- CelestialBody.h, CelestialBody.cpp (thread 381a8fe7 source)
- PAPER_171 (Ug1-Ug4 field functions that consume this struct)
- PAPER_172 (FU assembly)